

DASHBOARD DEVELOPMENT DOCUMENTATION

Project Title:

Retail & E-Commerce Sales Performance Analysis

1. Objective

The objective of this phase is to develop an interactive Power BI dashboard that enables users to analyze retail and e-commerce sales performance through key performance indicators (KPIs), charts, slicers, and drill-down features. The dashboard is designed to transform raw business data into meaningful visual insights that support data-driven decision-making.

2. Tool Used

- Microsoft Power BI
- DAX (Data Analysis Expressions)
- Power Query

3. Dashboard Overview

The Retail & E-Commerce Sales Dashboard provides an interactive view of business performance by analyzing sales, profit, orders, quantity sold, customer segments, regions, products, categories, payment methods, and returns. The dashboard allows users to filter information dynamically and quickly identify business trends and performance indicators.

4. Dashboard Components

4.1 KPI Cards

The dashboard contains the following Key Performance Indicators (KPIs):

- Total Sales
- Total Profit
- Total Orders
- Total Quantity Sold
- Returned Orders
- Return Rate

These KPIs provide a quick summary of overall business performance.

4.2 Interactive Slicers

The following slicers were added to allow users to filter the dashboard dynamically:

- Category
- Region

- Customer Segment
- Payment Method

Dropdown slicers were used to improve dashboard appearance and save report space.

4.3 Charts Used

The dashboard includes the following visualizations:

1. Monthly Sales Trend (Line Chart)

Purpose:

Analyze monthly sales performance and identify seasonal trends.

2. Sales & Profit by Category and Sub-Category (Drill-Down Column Chart)

Purpose:

Compare category performance and drill down into sub-categories for detailed analysis.

3. Sales by Product (Bar Chart)

Purpose:

Identify the highest-selling products.

4. Sales by Region (Column Chart)

Purpose:

Compare sales performance across different geographical regions.

5. Sales by Customer Segment (Donut Chart)

Purpose:

Analyze sales contribution by different customer segments.

6. Sales by Payment Method (Donut Chart)

Purpose:

Understand customer payment preferences.

5. Dashboard Features

The dashboard includes several interactive features to improve usability and business analysis.

- Interactive KPI Cards
- Dynamic Filtering using Slicers
- Category to Sub-Category Drill-Down
- Automatic Cross-Filtering between Visuals
- Interactive Data Exploration
- Responsive Dashboard Layout

6. Dashboard Design Principles

The dashboard was designed following standard data visualization best practices.

- Clean and simple layout
- Consistent color theme
- Proper alignment of visuals
- Equal spacing between charts
- Clearly labeled visuals
- Interactive navigation using slicers
- Business-friendly dashboard design

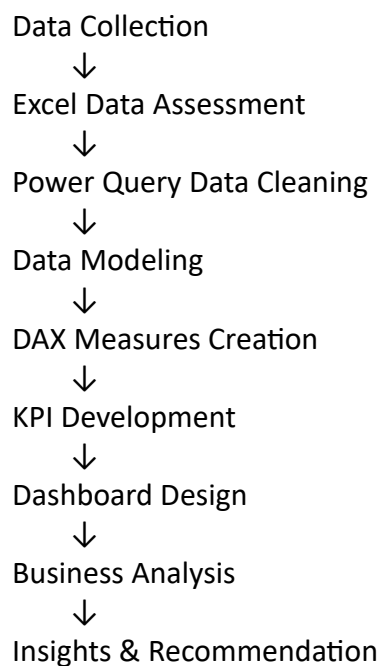
7. Business Benefits

The dashboard helps business users to:

- Monitor overall business performance.
- Identify top-performing products.
- Compare regional sales performance.
- Evaluate category and sub-category performance.
- Analyze customer purchasing behavior.
- Understand preferred payment methods.
- Track returned orders.
- Support strategic business decision-making.

8. Dashboard Workflow

The dashboard development followed the below workflow:



9. Conclusion

The Retail & E-Commerce Sales Dashboard successfully transforms raw transactional data into meaningful business insights using interactive visualizations. The combination of KPI cards, slicers, drill-down analysis, and dynamic charts enables users to monitor business performance efficiently. The dashboard supports data-driven decision-making by providing a clear understanding of sales trends, product performance, customer behavior, regional performance, payment preferences, and return analysis in a single interactive report.